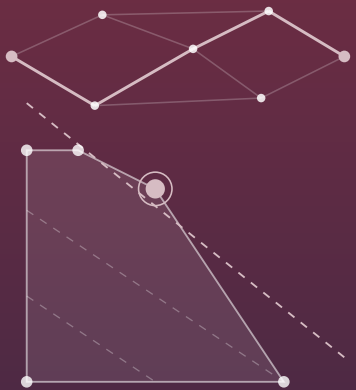
Operations Research

Lecture 3 · Semester 5



المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة

NHSAST

NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

The Simplex Method

Moving from corner to corner with algebra

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

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Operations Research (O.R.) · Semester 5 · 2026–2027

Today’s journey: six parts

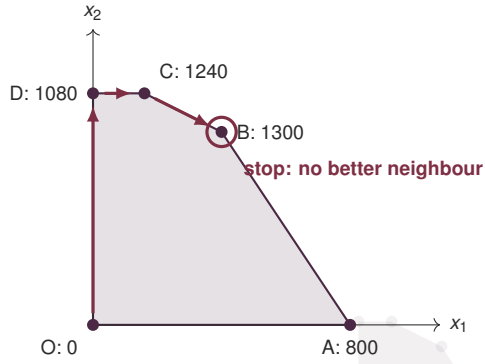
1	2	3	4	5	6
Corners as algebra	One step of the method	The full walk and the stop	The tableau	Three variables	Signals and mistakes

1. From corners to basic solutions
2. One step of the simplex method
3. The full walk, and when to stop
4. The simplex tableau
5. Three variables: no picture needed
6. Signals, why it works, and mistakes

Goal run the simplex method by hand, read the answer, and explain why it is the best.

Where we are: walk from corner to corner

- Lecture 2: if a best plan exists, a corner is optimal.
- Checking all corners is too slow: with 50 variables and 100 constraints, about 10^{40} systems to solve.
- Better idea: walk from a corner to a better neighbour. Stop when no neighbour is better.
- Today: this walk with algebra, without drawing.



$z = 40x_1 + 60x_2$ (thousand DA) at each corner

Part 1 of 6 · minutes 5–21

From corners to basic solutions

1. **Corners as algebra**
2. One step of the method
3. The full walk and the stop
4. The tableau
5. Three variables
6. Signals and mistakes

In this part, you learn to

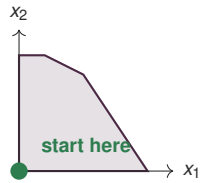
- describe a corner with numbers: which variables are 0;
- compute a basic solution and check if it is feasible;
- recognise two neighbouring corners.

What we assume today

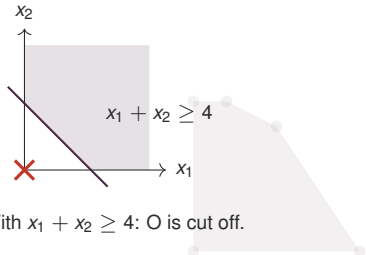
Today's problems

$\max\ z = c^\top x \quad \text{s.t.} \quad Ax \le b,\ x \ge 0, \quad \text{with } b \ge 0.$

- Then $x = 0$ is feasible ($A \cdot 0 = 0 \le b$), and it is a corner: we start at the origin O.
- Drone workshop: $b = (40, 60, 18) \ge 0$ ✓
- To minimise z: maximise $-z$ (Lecture 1).
- With "$\ge$", "$=$" or some $b_i < 0$, the origin can be outside the feasible set. We start in another way: Lecture 4.



Drone workshop: O is feasible.



With $x_1 + x_2 \ge 4$: O is cut off.

Slack variables: the part we do not use

Drone workshop in standard form (Lecture 1):

$$\max z = 40x_1 + 60x_2$$

$$x_1 + 2x_2 + s_1 = 40 \quad \text{assembly}$$

$$3x_1 + 2x_2 + s_2 = 60 \quad \text{testing}$$

$$x_2 + s_3 = 18 \quad \text{cameras}$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Plan P $= (6, 10)$**: inside**



Corner B $= (10, 15)$

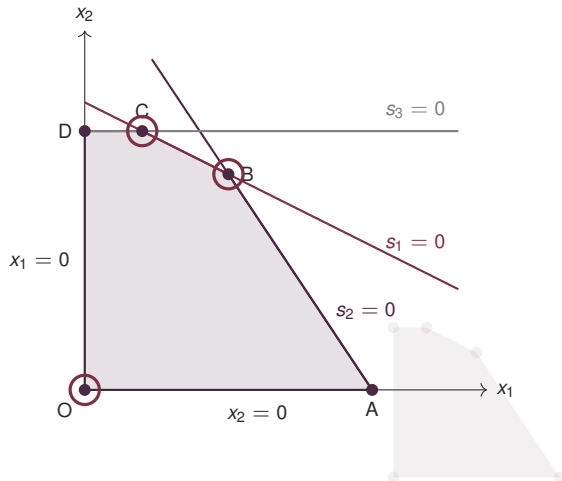


$s_i =$ the part of resource i that is not used.

Remember slack $= 0 \Leftrightarrow$ resource fully used $\Leftrightarrow$ the point is on that line.

At a corner, two variables are zero

- 5 variables, 3 equations.
- On each line, its variable is 0.
- Corner B is on two lines: $s_1 = 0$ and $s_2 = 0$.
- Corner C: $s_1 = 0$ and $s_3 = 0$.
- Corner O: $x_1 = 0$ and $x_2 = 0$.
- At each corner: 2 of the 5 variables are 0. The 3 others come from the 3 equations.



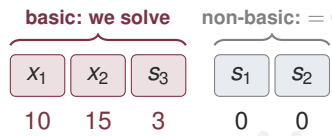
Definition 3.1: basic solution

Definition 3.1 (basic solution)

A system has m equations. Choose m variables: the basic variables. The other variables are non-basic: put them equal to 0. Solve the equations for the basic variables: this gives a basic solution. If all its values are ≥ 0, it is a basic feasible solution (BFS).

Example ($m = 3$): non-basic s_1, s_2.

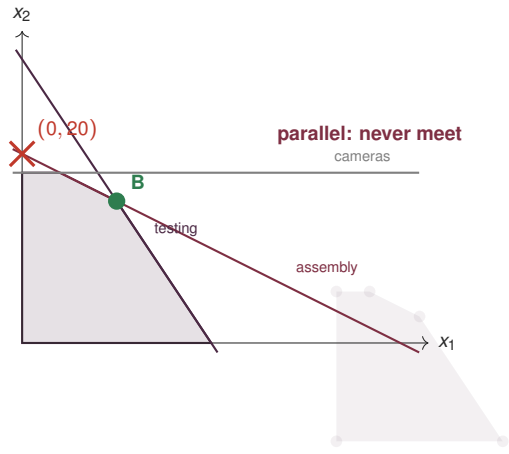
1. Put $s_1 = s_2 = 0$: $x_1 + 2x_2 = 40$, $3x_1 + 2x_2 = 60$, $x_2 + s_3 = 18$.
2. Second minus first: $2x_1 = 20$, so $x_1 = 10$; then $x_2 = 15$ and $s_3 = 3$.
3. $(10, 15, 0, 0, 3)$: all ≥ 0 ✓ BFS: the corner B.



Three choices, three results

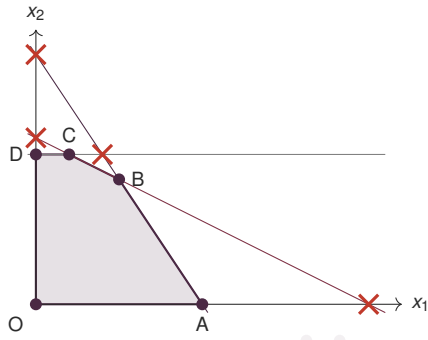
1. Non-basic s_1, s_2: assembly and testing lines meet at $(10, 15)$; $s_3 = 3$. **Feasible: corner B** ✓
2. Non-basic x_1, s_1: the x_2-axis and the assembly line meet at $(0, 20)$; $s_3 = 18 - 20 = -2$. **Not feasible** ×
3. Non-basic x_2, s_3: the lines $x_2 = 0$ and $x_2 = 18$ are parallel. **No solution.**

Picture two lines meet = basic solution;
in the region = feasible.



Key fact 3.1: feasible basic solutions = corners

Non-basic ($= 0$)	$(x_1, x_2, s_1, s_2, s_3)$	Point
x_1, x_2	$(0, 0, 40, 60, 18)$	O
x_2, s_2	$(20, 0, 20, 0, 18)$	A
s_1, s_2	$(10, 15, 0, 0, 3)$	B
s_1, s_3	$(4, 18, 0, 12, 0)$	C
x_1, s_3	$(0, 18, 4, 24, 0)$	D
x_1, s_1	$(0, 20, 0, 20, -2)$	×
x_1, s_2	$(0, 30, -20, 0, -12)$	×
x_2, s_1	$(40, 0, 0, -60, 18)$	×
s_2, s_3	$(8, 18, -4, 0, 0)$	×
x_2, s_3	no solution (parallel)	—



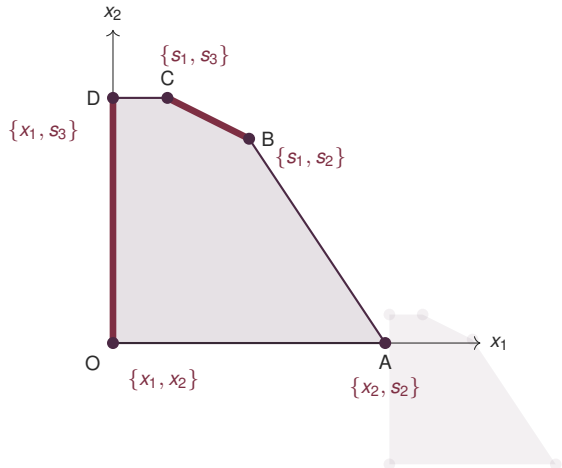
Key fact 3.1

The basic feasible solutions are exactly the corners of the feasible set.

Neighbouring corners: one in, one out

- Each corner has its pair of zeros (non-basic variables).
- Corners joined by an edge are neighbours: their pairs share one variable.
- O $\{x_1, x_2\} \to$ D $\{x_1, s_3\}$: x_2 stops being 0, it enters; s_3 becomes 0, it leaves.
- C $\{s_1, s_3\} \to$ B $\{s_1, s_2\}$: s_3 enters, s_2 leaves.

One step one variable enters, one variable leaves.



Part 2 of 6 · minutes 21–37

One step of the simplex method

1 Corners as algebra — 2 **One step of the method** — 3 The full walk and the stop — 4 The tableau — 5 Three variables — 6 Signals and mistakes

In this part, you learn to

- write the dictionary of a corner;
- choose the variable that enters (who grows?);
- do the ratio test (how far?) and the pivot (the new corner).

The dictionary of the corner O

Isolate each basic variable (here: the slacks):

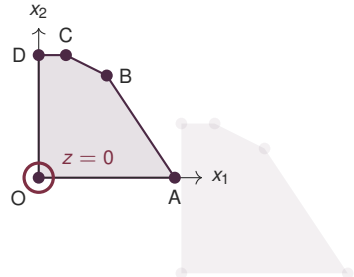
$$
\begin{aligned}
x_1 + 2x_2 + s_1 = 40 \quad &\Rightarrow \quad s_1 = 40 - x_1 - 2x_2 \\
3x_1 + 2x_2 + s_2 = 60 \quad &\Rightarrow \quad s_2 = 60 - 3x_1 - 2x_2 \\
x_2 + s_3 = 18 \quad &\Rightarrow \quad s_3 = 18 \qquad - x_2 \\
\hline
z = 40x_1 + 60x_2 \quad &\Rightarrow \quad z \;=\; 0 + 40x_1 + 60x_2
\end{aligned}
$$

Read it: non-basic $x_1 = x_2 = 0 \Rightarrow s_1 = 40$, $s_2 = 60$, $s_3 = 18$, $z = 0$.

This is the corner O: nothing made, nothing used.

Definition 3.2 (dictionary)

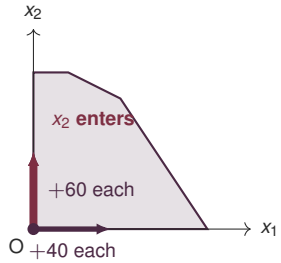
The basic variables and z, written with the non-basic variables only.



Question 1: who enters?

$$z = 0 + 40\,x_1 + 60\,x_2$$

- Now nothing is made: $x_1 = x_2 = 0$.
- One more drone A: z grows by 40.
  One more drone B: z grows by 60.
- A positive number in the z-line: this variable can improve z.
- We take the largest: x_2 enters.



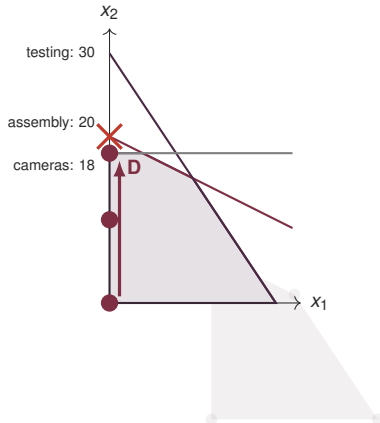
Rule 3.1 (entering variable)

Choose a non-basic variable with a positive number in the z-line (usually the largest). If there is none, the corner is optimal (Key fact 3.2).

Question 2: how far can x_2 grow? (ratio test)

Keep $x_1 = 0$ and let $x_2 = t$ grow:

t	$s_1 = 40 - 2t$	$s_2 = 60 - 2t$	$s_3 = 18 - t$	
0	40	60	18	start: O
10	20	40	8	inside
18	4	24	0	stop here
20	0	20	-2	too far!



Ratio test

s_1: $t \le 40/2 = 20$; s_2: $t \le 60/2 = 30$;

s_3: $t \le 18/1 = 18$ $\leftarrow$ the smallest ratio.

So $x_2 = 18$, and s_3 leaves (it becomes 0).

Question 3: the pivot (the new dictionary)

1. From the line of s_3 (it leaves):

$s_3 = 18 - x_2 \Rightarrow x_2 = 18 - s_3$.

2. Put it in the other lines:

$s_1 = 40 - x_1 - 2(18 - s_3) = 4 - x_1 + 2s_3$

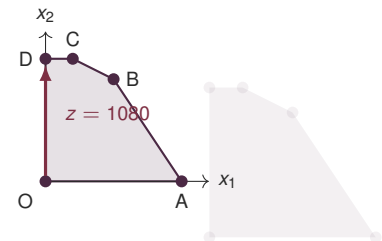
$s_2 = 60 - 3x_1 - 2(18 - s_3) = 24 - 3x_1 + 2s_3$

$z = 40x_1 + 60(18 - s_3) = 1080 + 40x_1 - 60s_3$

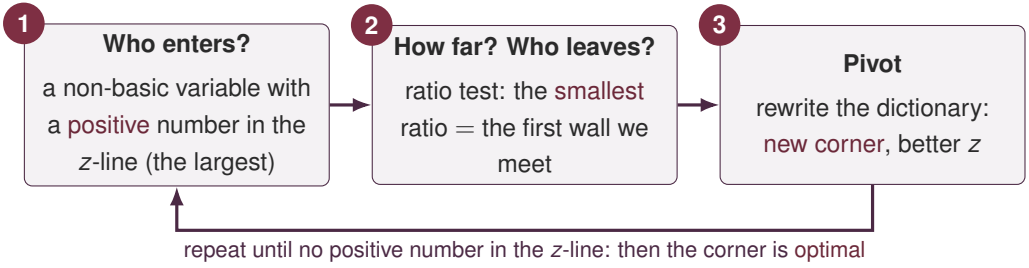
3. Read it: $x_1 = s_3 = 0 \Rightarrow x_2 = 18$, $s_1 = 4$, $s_2 = 24$, $z = 1080$: the corner D.

New dictionary (non-basic: x_1, s_3)

$$x_2 = 18 - s_3$$
$$s_1 = 4 - x_1 + 2s_3$$
$$s_2 = 24 - 3x_1 + 2s_3$$
$$z = 1080 + 40x_1 - 60s_3$$



One step of the simplex method = three questions



Our first step: x_2 enters $\rightarrow$ s_3 leaves (ratio 18) $\rightarrow$ corner D, z: $0 \rightarrow 1080$.

Part 3 of 6 · minutes 37–52

The full walk, and when to stop

1 Corners as algebra — 2 One step of the method — **3 The full walk and the stop** — 4 The tableau — 5 Three variables — 6 Signals and mistakes

In this part, you learn to

- finish the drone workshop: D $\to$ C $\to$ B;
- understand a surprising step: a slack enters;
- explain why we can stop: the last z-line is a proof.

Second step: x_1 enters

Dictionary at D:

$$x_2 = 18 - s_3, \qquad s_1 = 4 - x_1 + 2s_3,$$
$$s_2 = 24 - 3x_1 + 2s_3, \qquad z = 1080 + 40\,x_1 - 60\,s_3.$$

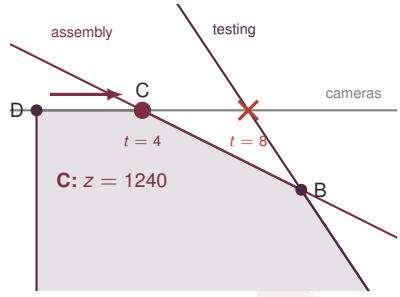
1. Who enters? Only x_1 has a positive number: x_1 enters.

2. How far? ($s_3 = 0$, $x_1 = t$)

$s_1 = 4 - t \geq 0$: $t \leq 4$; $\quad s_2 = 24 - 3t \geq 0$: $t \leq 8$; $x_2 = 18$: no limit.

Smallest: $4 \Rightarrow s_1$ leaves.

3. Pivot: $x_1 = 4 - s_1 + 2s_3$, substitute $\Rightarrow$ corner C $= (4, 18)$, $z = 1240$.



zoom on the top of the picture

Third step: a surprise — s_3 enters

Dictionary at C:

$$x_1 = 4 - s_1 + 2s_3$$
$$s_2 = 12 + 3s_1 - 4s_3$$
$$x_2 = 18 - s_3$$
$$z = 1240 - 40s_1 + 20s_3$$

s_3 has a positive number: s_3 enters.

But s_3 = unused cameras! Why leave cameras unused?

Keep assembly full ($s_1 = 0$).
Leave one camera unused ($s_3 = 1$).

Line x_2: $x_2 = 18 - 1$.
One drone B less: -60.
It frees 2 assembly hours.

Line x_1: $x_1 = 4 + 2$.
Two drones A more: $+80$.

$-60 + 80 = +\mathbf{20}$ per unused camera:
the 20 in the z-line ✓

Third step: how far? The corner B

2. How far? ($s_1 = 0$, $s_3 = t$)

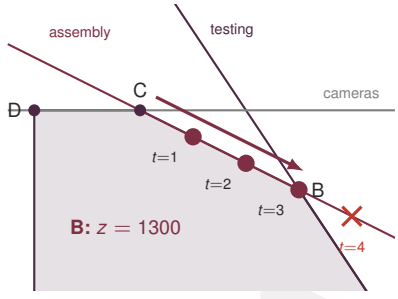
- $x_1 = 4 + 2t$: grows, no limit;
- $s_2 = 12 - 4t \geq 0$: $t \leq 12/4 = 3$;
- $x_2 = 18 - t \geq 0$: $t \leq 18$.

Smallest: $3 \Rightarrow s_2$ leaves.

3. Pivot: from the line of s_2: $s_3 = 3 + \frac{3}{4}s_1 - \frac{1}{4}s_2$, then

$$x_1 = 10 + \tfrac{1}{2}s_1 - \tfrac{1}{2}s_2, \qquad x_2 = 15 - \tfrac{3}{4}s_1 + \tfrac{1}{4}s_2,$$
$$z = 1300 - 25s_1 - 5s_2.$$

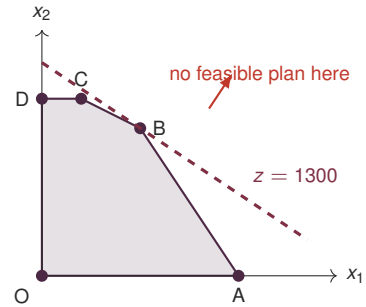
Corner B $= (10, 15)$, $z = 1300$.



zoom on the top of the picture

Stop: B is optimal — and here is the proof

1. $z = 1300 - 25\,s_1 - 5\,s_2$: no positive number. Stop.
2. This line is true for every feasible plan (the equations, rewritten).
3. Every feasible plan has $s_1 \geq 0$ and $s_2 \geq 0$.
4. So $z \leq 1300$, and B gives 1300. ✓
5. 25 and 5 = the multipliers of Lecture 2.



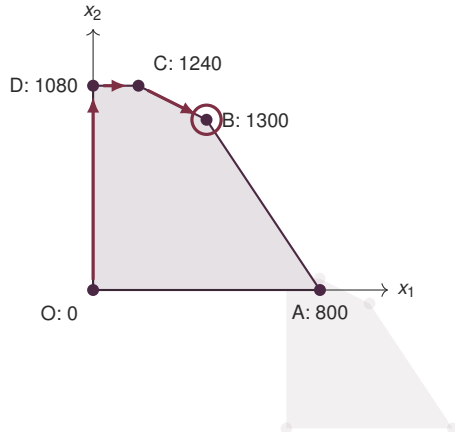
Key fact 3.2 (optimality)

No positive number in the z-line $\Rightarrow$ the corner is optimal; its value is the constant of the z-line.

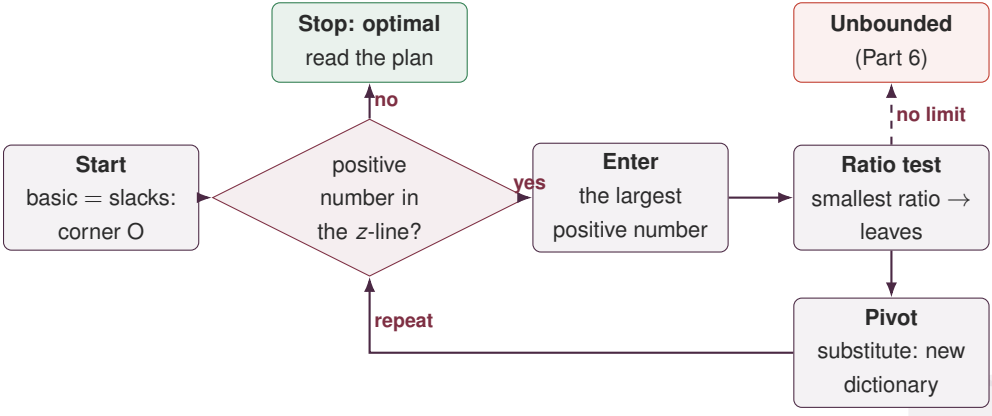
The whole walk

Step	Enters	Leaves	Corner	z
start	—	—	O $(0, 0)$	0
1	x_2	s_3	D $(0, 18)$	1080
2	x_1	s_1	C $(4, 18)$	1240
3	s_3	s_2	B $(10, 15)$	1300

- Each step: one edge, to a better neighbour.
- 4 corners visited, no drawing needed.
- If x_1 enters first: O $\rightarrow$ A $\rightarrow$ B (TD 2).



Method 3.1: the simplex method with dictionaries



Ratio test: only lines where the entering variable has a negative coefficient; ratio = constant ÷ size of the coefficient.

Part 4 of 6 · minutes 52–67

The simplex tableau

1 Corners as algebra — 2 One step of the method — 3 The full walk and the stop — **4 The tableau** — 5 Three variables — 6 Signals and mistakes

In this part, you learn to

- write the same equations in a table: the tableau;
- find the pivot in the table;
- do the row operations and read the new corner.

From the dictionary to the tableau

dictionary

$$
\begin{aligned}
s_1 &= 40 - x_1 - 2x_2 \\
s_2 &= 60 - 3x_1 - 2x_2 \\
s_3 &= 18 \qquad\quad - x_2 \\
\hline
z &= \quad 0 + 40x_1 + 60x_2
\end{aligned}
$$

variables to the left →

tableau

T0	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	2	1	0	0	40
s_2	3	2	0	1	0	60
s_3	0	1	0	0	1	18
z	−40	−60	0	0	0	0

- Each line is an equation: line s_1 reads $x_1 + 2x_2 + s_1 = 40$.
- The z-line reads $z - 40x_1 - 60x_2 = 0$: the signs change.
- The basic variables have unit columns: a 1 in their line, 0 elsewhere (also in the z-line).
- RHS = the values of the basic variables, and (last line) the value of z.

Rule 3.2: the three questions in the tableau

	Dictionary	Tableau
1. Who enters?	largest positive number in the z-line	most negative number in the z-line
2. How far?	lines with a negative coefficient: constant $\div$ size	lines with a positive number: RHS $\div$ number; smallest ratio
3. Pivot	solve the leaving line, substitute	divide the pivot line by the pivot; make 0 in the rest of the column (row operations)
Stop when	no positive number	no negative number

Why? the tableau writes $z - 40x_1 - 60x_2 = 0$: same rules, opposite signs.

Pivot 1 in the tableau: corner O → D

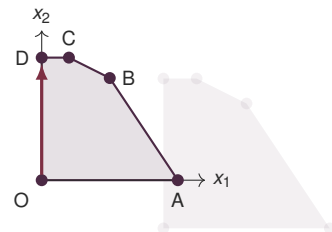
enters ↓

T0	x_1	x_2	s_1	s_2	s_3	RHS	ratio
s_1	1	2	1	0	0	40	$40/2 = 20$
s_2	3	2	0	1	0	60	$60/2 = 30$
leaves s_3	0	1	0	0	1	18	$18/1 = 18$
z	-40	-60	0	0	0	0	

pivot

T1	x_1	x_2	s_1	s_2	s_3	RHS	
s_1	1	0	1	0	-2	4	$L_1 - 2L_3$
s_2	3	0	0	1	-2	24	$L_2 - 2L_3$
x_2	0	1	0	0	1	18	$L_3 \div 1$
z	-40	0	0	0	60	1080	$z + 60L_3$

1. Most negative: -60: x_2 enters.
2. Ratios: RHS $\div$ positive numbers.
3. Smallest 18: s_3 leaves; pivot 1.
4. Row operations (right of T1).
5. T1: corner D, $z = 1080$.



Pivot 2 in the tableau: corner D → C

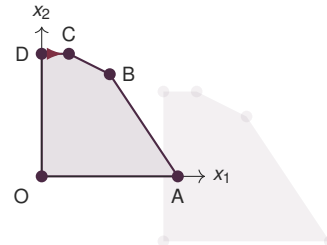
enters ↓

T1	x_1	x_2	s_1	s_2	s_3	RHS	ratio
leaves s_1	1	0	1	0	−2	4	$4/1 = 4$
s_2	3	0	0	1	−2	24	$24/3 = 8$
x_2	0	1	0	0	1	18	0: no ratio
z	−40	0	0	0	60	1080	

pivot

T2	x_1	x_2	s_1	s_2	s_3	RHS	
x_1	1	0	1	0	−2	4	$L_1 \div 1$
s_2	0	0	−3	1	4	12	$L_2 - 3L_1$
x_2	0	1	0	0	1	18	unchanged
z	0	0	40	0	−20	1240	$z + 40L_1$

1. Only negative: -40: x_1 enters.
2. Line x_2 has a 0: no ratio.
3. Smallest 4: s_1 leaves; pivot 1.
4. T2: corner C, $z = 1240$.
5. Still -20 in the z-line: go on.



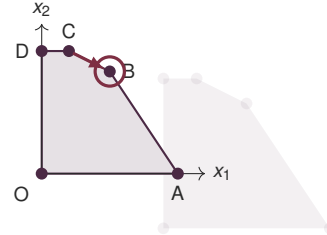
Pivot 3 in the tableau: corner C → B, optimal

						enters ↓		
	T2	x_1	x_2	s_1	s_2	s_3	RHS	ratio
	x_1	1	0	1	0	-2	4	-2: no ratio
leaves	s_2	0	0	-3	1	4	12	$12/4 = 3$
	x_2	0	1	0	0	1	18	$18/1 = 18$
	z	0	0	40	0	-20	1240	

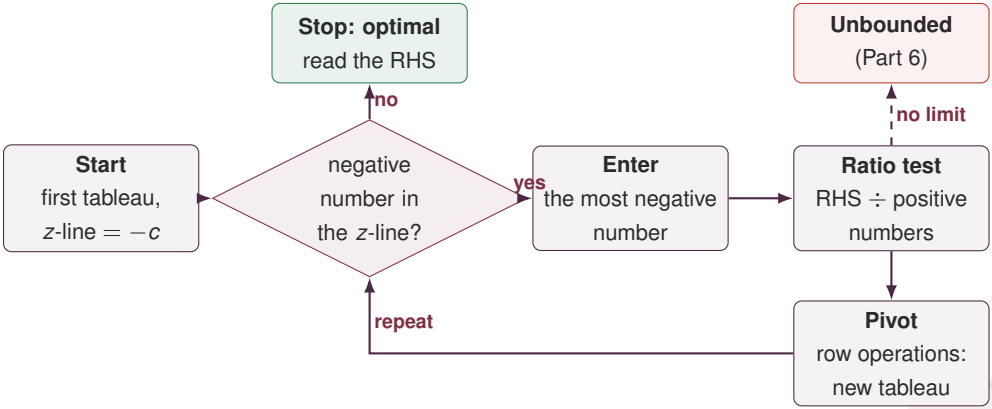
pivot

	T3	x_1	x_2	s_1	s_2	s_3	RHS	
	x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	10	$L_1 + 2L_2$
	s_3	0	0	$-\frac{3}{4}$	$\frac{1}{4}$	1	3	$L_2 \div 4$
	x_2	0	1	$\frac{3}{4}$	$-\frac{1}{4}$	0	15	$L_3 - L_2$
	z	0	0	25	5	0	1300	$z + 20L_2$

1. -20: s_3 enters.
2. Line x_1 has -2: no ratio.
3. Smallest 3: s_2 leaves; pivot 4: divide L_2 by 4 first.
4. T3: no negative number: optimal. Corner B, $z = 1300$.



Method 3.2: the simplex method with tableaux



Other books write the last line with the opposite signs ($c_j - z_j$): then choose the largest positive number, and stop when all are ≤ 0. Same method.

Part 5 of 6 · minutes 67–76

Three variables: no picture needed

1 Corners as algebra — 2 One step of the method — 3 The full walk and the stop — 4 The tableau — **5 Three variables** — 6 Signals and mistakes

In this part, you learn to

- run the tableau on a problem we cannot draw;
- read the plan and the unused resources;
- check the answer in the original constraints.

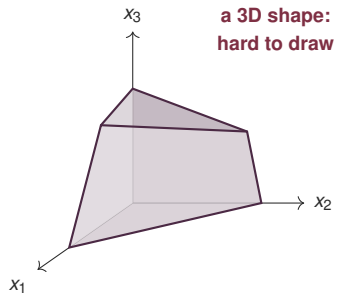
Example 3.2: three robot kits

A workshop builds three robot kits. x_j = number of kits Kj per day.

	K1	K2	K3	Per day
Cutting (h per kit)	3	2	2	24
Assembly (h per kit)	2	2	2	30
Testing (h per kit)	3	1	2	20
Profit (thousand DA)	3	3	5	

$$\max z = 3x_1 + 3x_2 + 5x_3$$

$$\text{s.t. } 3x_1 + 2x_2 + 2x_3 \le 24, \quad 2x_1 + 2x_2 + 2x_3 \le 30,$$
$$3x_1 + x_2 + 2x_3 \le 20, \quad x_1, x_2, x_3 \ge 0$$



a 3D shape: hard to draw

Your turn · 2 min write the first tableau.

Example 3.2: pivot 1

enters ↓

T0	x_1	x_2	x_3	s_1	s_2	s_3	RHS	ratio
s_1	3	2	2	1	0	0	24	$24/2 = 12$
s_2	2	2	2	0	1	0	30	$30/2 = 15$
leaves s_3	3	1	2	0	0	1	20	$20/2 = 10$
z	-3	-3	-5	0	0	0	0	

pivot

T1	x_1	x_2	x_3	s_1	s_2	s_3	RHS	
s_1	0	1	0	1	0	-1	4	$L_1 - 2L_3$
s_2	-1	1	0	0	1	-1	10	$L_2 - 2L_3$
x_3	$\frac{3}{2}$	$\frac{1}{2}$	1	0	0	$\frac{1}{2}$	10	$L_3 \div 2$
z	$\frac{9}{2}$	$-\frac{1}{2}$	0	0	0	$\frac{5}{2}$	50	$z + 5L_3$

1. Most negative: $-5 \Rightarrow x_3$ enters (kit K3).
2. Ratios: 12, 15, 10.
3. Smallest: $10 \Rightarrow s_3$ leaves (testing). Pivot $= 2$: divide L_3 by 2 first.
4. T1: $z = 50$. Still $-\frac{1}{2}$ under x_2: continue.

Example 3.2: pivot 2, optimal

enters ↓

	T1	x_1	x_2	x_3	s_1	s_2	s_3	RHS	ratio
leaves	s_1	0	1	0	1	0	-1	4	$4/1 = 4$
	s_2	-1	1	0	0	1	-1	10	$10/1 = 10$
	x_3	$\frac{3}{2}$	$\frac{1}{2}$	1	0	0	$\frac{1}{2}$	10	$10/\frac{1}{2} = 20$
	z	$\frac{9}{2}$	$-\frac{1}{2}$	0	0	0	$\frac{5}{2}$	50	

pivot ↓

T2	x_1	x_2	x_3	s_1	s_2	s_3	RHS	
x_2	0	1	0	1	0	-1	4	$L_1 \div 1$
s_2	-1	0	0	-1	1	0	6	$L_2 - L_1$
x_3	$\frac{3}{2}$	0	1	$-\frac{1}{2}$	0	1	8	$L_3 - \frac{1}{2}L_1$
z	$\frac{9}{2}$	0	0	$\frac{1}{2}$	0	2	52	$z + \frac{1}{2}L_1$

1. $-\frac{1}{2} \Rightarrow x_2$ enters.
2. Ratios: 4, 10, and $10/\frac{1}{2} = 20$.
3. Smallest: $4 \Rightarrow s_1$ leaves. Pivot $= 1$.
4. T2: no negative number: optimal, $z = 52$.

Example 3.2: reading the answer

- Basic: $x_2 = 4$, $s_2 = 6$, $x_3 = 8$. Non-basic: $x_1 = s_1 = s_3 = 0$.
- Plan: 0 kits K1, 4 kits K2, 8 kits K3 per day. Profit 52 thousand DA.
- Cutting and testing fully used ($s_1 = s_3 = 0$); 6 assembly hours unused ($s_2 = 6$).
- Why no K1? Its number in the z-line is $\frac{9}{2}$: one kit K1 would lower z by 4.5.



Always check

Cutting: $0 + 8 + 16 = 24 \leq 24$ ✓
Assembly: $0 + 8 + 16 = 24 \leq 30$ ✓
Testing: $0 + 4 + 16 = 20 \leq 20$ ✓
Profit: $0 + 12 + 40 = 52$ ✓

Part 6 of 6 · minutes 76–88

Signals, why it works, and mistakes

1 Corners as algebra
2 One step of the method
3 The full walk and the stop
4 The tableau
5 Three variables
6 **Signals and mistakes**

In this part, you learn to

- recognise an unbounded problem in the tableau;
- recognise many optimal plans in the tableau;
- explain why the method works, and avoid the common mistakes.

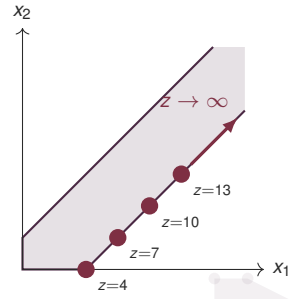
Signal 1: no ratio $\rightarrow$ the problem is unbounded

max $z = 2x_1 + x_2$ s.t. $x_1 - x_2 \le 2$, $-x_1 + x_2 \le 1$, $x_1, x_2 \ge 0$. After one pivot:

enters $\downarrow$

T1	x_1	x_2	s_1	s_2	RHS	ratio
x_1	1	-1	1	0	2	-1: no ratio
s_2	0	0	1	1	3	0: no ratio
z	0	-3	2	0	4	

Put $x_2 = t$, $s_1 = 0$: $x_1 = 2 + t$, $s_2 = 3$,
$z = 4 + 3t \rightarrow +\infty$.



Rule 3.3 (unbounded problem)

If the entering column has no positive number, the problem is unbounded: z grows without limit.

Signal 2: a 0 in the z-line $\rightarrow$ many optimal plans

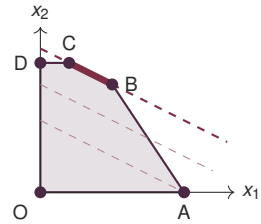
Drone workshop, but drone A earns only 30: $\max\ z = 30x_1 + 60x_2$. After two pivots (corner C):

0: try it ↓

	T2	x_1	x_2	s_1	s_2	s_3	RHS	ratio
	x_1	1	0	1	0	-2	4	-2: no ratio
leaves	s_2	0	0	-3	1	4	12	$12/4 = 3$
	x_2	0	1	0	0	1	18	$18/1 = 18$
	z	0	0	30	0	0	1200	

pivot (z stays 1200)

T3	x_1	x_2	s_1	s_2	s_3	RHS	
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	10	$L_1 + 2L_2$
s_3	0	0	$-\frac{3}{4}$	$\frac{1}{4}$	1	3	$L_2 \div 4$
x_2	0	1	$\frac{3}{4}$	$-\frac{1}{4}$	0	15	$L_3 - L_2$
z	0	0	30	0	0	1200	unchanged

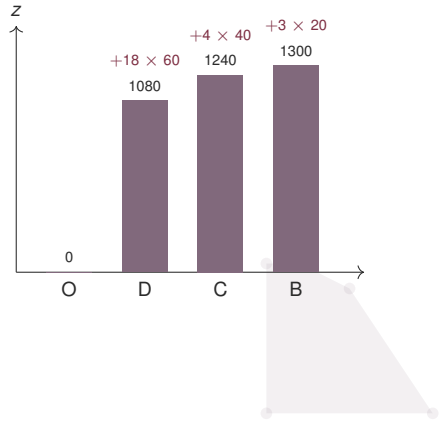


- T2 optimal: corner C, $z = 1200$.
- But 0 under non-basic s_3: z will not change.
- Corner B, also $z = 1200$: the edge C–B is optimal.

Why the simplex method works

1. z **never goes down.** Each pivot adds (ratio) $\times$ (number of the entering variable) ≥ 0.
2. **It stops.** z grows at each pivot, so no corner is visited twice; and there are finitely many corners.
3. **The stop is a proof.** The last z-line gives $z \leq z^*$ for every feasible plan (Key fact 3.2).

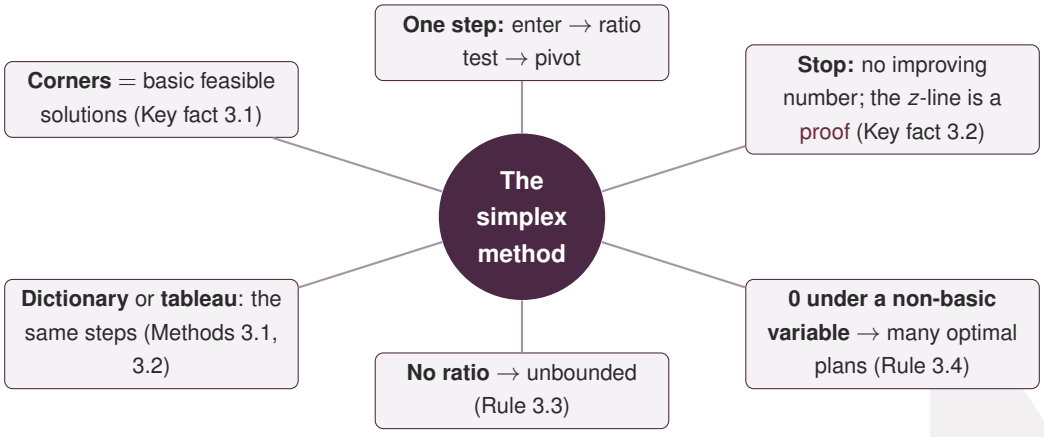
One exception: a ratio 0 gives no growth (a *degenerate* corner). This, and starting when O is not feasible: Lecture 4.



Common mistakes

Wrong ×	Right ✓
A ratio with a 0 or negative number	Only positive numbers of the column
The largest ratio, or the largest number	Always the smallest ratio
Divide only the pivot	Divide the whole pivot line, RHS too
Forget the z-line	The z-line is a line like the others
A negative RHS appears	There is a mistake: wrong pivot or arithmetic
Only numbers in the answer	Plan, unused resources, z, and a check

Summary: the simplex method in one picture



TD 2: the picture, then the simplex method. Next: Lecture 4 (origin not feasible; degeneracy).